

Course: B.Pharm.

Semester: IV

Subject with Subject Code: Pharmaceutical Engineering (BP304T)

Date: 26/12/2018

Marks: 75

Duration: 3hrs

Instructions: i) All questions are compulsory

ii) Figures to the right indicate full marks

iii) Draw the diagrams or flow charts wherever necessary.

Q.No.1 Attempt the following questions (All Questions Compulsory)

(20 Marks)

- Austenitic consists of
 - 13 to 20% Chromium + 6 to 22% Nickel + 0.1 to 0.25% carbon
 - 12 to 20% Chromium + 2% Nickel + 0.2 to 0.4% carbon
 - 20 to 40% Chromium + 12% Nickel + 1 to 2% carbon
 - 15 to 30% Chromium + 0.1% carbon
- Commonly used ball for pebble mill are
 - i) Round ball,
 - ii) Rods
 - iii) Needle
 - Only i are used
 - All ii & iii are used
 - i & ii are used
 - All of these are used.
- Which of the following is false for reflux ratio for High efficiency of fractional distillation?
 - It is controlled by means of a suitable still.
 - It should be low.
 - The quotient of the amount of liquid returning through the column to the amount collected into the receiver during the same interval of time.
 - It should be high.
- As per Indian official standard Moderately fine powder is
 - All particles must pass through sieve no 44 and 50 % particles pass through sieve no 85.
 - All particles must pass through sieve no 44 and 60 % particles pass through sieve no 85.
 - All particles must pass through sieve no 44 and 50 % particles pass through sieve no 60.
 - All particles must pass through sieve no 44 and 40 % particles pass through sieve no 85.
- Free moisture content is
 - Total water content minus equilibrium moisture content.
 - Total water content plus equilibrium moisture content.
 - Ratio of Total water content to the equilibrium moisture content.
 - Total water present in solid minus water in environment.

6. The Liquid mixing mechanism are
 - A. Bulk transport
 - B. Turbulent mixing
 - C. Molecular diffusion
 - D. All of these
7. The value of Reynold's number for turbulent flow is
 - A. <2000
 - B. 2000-4000
 - C. >4000
 - D. <1000
8. Ultracentrifuge shows
 - A. 85000 revolutions per minutes
 - B. 8500 revolutions per minutes
 - C. 10000 revolutions per minutes
 - D. 850 revolutions per minutes
9. Which of following is not advantage of rising film evaporator?
 - A. The enhanced heat transfer.
 - B. Suitable for foam forming liquids.
 - C. Large area for heat transfer is provided.
 - D. Not suitable for heat sensitive material.
10. In centrifuges the driving forces for separation of solids are
 - A. Centrifugal force
 - B. Both Centrifugal force & Gravitational force
 - C. Gravitational Force
 - D. None of these.
11. As per Fourier's law of convention rate of heat transfer through a uniform material is
 - A. Directly proportional to the length of uniform material.
 - B. Inversely proportional to the temperature difference.
 - C. Inversely proportional to the area of uniform material.
 - D. Directly proportional to the temperature drop.
12. Planetary mixer is are example of
 - A. Agitator mixer
 - B. Solid mixer
 - C. Shear mixer
 - D. All of these
13. Impingement, Entanglement & Straining are related to
 - A. Mixing
 - B. Centrifugation
 - C. Filtration
 - D. All of these
14. The conveyers for transportation of solids are
 - A. Belt conveyors
 - B. Chain conveyors
 - C. Screw conveyors
 - D. All of these
15. The bacteria used to test membrane filters of Pore size 0.3 (μm) are
 - A. *Serratia marcescens*
 - B. *Pseudomonas aeruginosa*
 - C. *Pseudomonas diminuta*
 - D. *Saccharomyces cerevisiae*
16. Ball mill shows the principle
 - A. Impact
 - B. Compression
 - C. Impact & attrition
 - D. Crushing & sharing
17. Mixing device technically called as
 - A. Impellers
 - B. Turbines
 - C. Paddles
 - D. All of these.

18. The rate of evaporation is
- Inversely proportional to atmospheric pressure
 - Inversely proportional to temperature.
 - Inversely proportional to the vapour pressure of liquid.
 - Inversely proportional to the surface area for evaporation.
19. Tunnel dryer is variant of
- Rotary drum dryer.
 - Fluidized bed dryer.
 - Tray dryer.
 - Spray dryer.
20. Raoult's law is related to
- Vapour pressure
 - Atmospheric pressure
 - Osmotic pressure
 - All the above

Q.No.2 Attempt any TWO questions of the following: (20 Marks)

- Derive Bernoulli's equation. Discuss its applications.
- Discuss in detail various modes of heat transfer. Draw a neat diagram of shell & tube heat exchanger & explain its working.
- List the factors influencing the rate of filtration. Explain construction, working & applications of filter press with a neat diagram.

Q.No.3 Attempt any SEVEN questions of the following: (35 Marks)

- Explain theories of corrosion.
- Discuss construction, working, application & advantages of fluidized bed dryer with a neat labeled diagram.
- Draw a neat diagram of bag filter & explain its working.
- Classify evaporators and explain economy of multiple effect evaporators in comparison to single effect evaporator.
- List the equipments used for solid mixing in pharmaceutical industry. Explain construction & working of Sigma blade mixer.
- Discuss the principle & application of centrifugation.
- Discuss any five factors affecting evaporation.
- With help of a neat diagram explain construction, working, application & advantages of fluid energy mill.
- What is meant by steam distillation? What are its special advantages?

----- END OF PAPER -----



**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103**

Winter Semester Examination - Dec. - 2019

Course: B. Pharm.

Subject: Pharmaceutical Engineering (BP304T)

Date: 24/12/2019

Semester: III

Marks: 75

Duration: 3hrs

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Attempt the following questions (All Questions Compulsory) (20)

- 1) The SI unit of Reynolds number is
 - a) Nm^{-2}
 - b) Centipoise
 - c) m/s
 - d) Unit less
- 2) Bernoulli's equation can be derived from the conservation of
 - a) Energy
 - b) Mass
 - c) Volume
 - d) Angular momentum
- 3) Fluid energy mill works on the principle of
 - a) Impact
 - b) Attrition
 - c) Cutting
 - d) Impact & attrition
- 4) Elutriation is a process of
 - a) Size reduction by mechanical forces
 - b) Size reduction by stationary fluid
 - c) Size reduction by moving fluid
 - d) Size reduction by electrical forces
- 5) Which of the following is the case of heat transfer by radiation
 - a) Blast furnace
 - b) Heating and building
 - c) Cooling parts of furnace
 - d) Heat received by a person from fire
- 6) Evaporation occurs only at
 - a) boiling
 - b) Extreme cooling
 - c) Surface of a liquid
 - d) Boiling at atmospheric pressure
- 7) Sieve number indicates, the number of meshes in
 - a) 2.54 mm
 - b) 25.4 mm
 - c) 254 mm
 - d) 0.254 mm
- 8) Vena contracta occurs in
 - a) Venturi meter
 - b) Orifice meter
 - c) Pitot tube
 - d) Rota meter

- 9) Which of the following equation contains the term, "Specific surface area"
- Darcy's equation
 - Kozeny-carman equation
 - Stock's law
 - Poiseuille's equation
- 10) Fluidization as a concept involved in
- Drying
 - Coating
 - Atomizers
 - All of the above
- 11) Which distillation is called as 'Evaporative distillation' or 'Short path distillation'?
- Simple distillation
 - Vacuum distillation
 - Molecular distillation
 - Fractional distillation
- 12) Differential distillation is?
- Simple distillation
 - Vacuum distillation
 - Molecular distillation
 - Fractional distillation
- 13) In which of the following techniques the sample is kept below triple point
- Centrifugation
 - Spray congealing
 - Lyophilization
 - Spray Drying
- 14) If the dry spot appears in the substance in the batch drying curve at?
- CMC
 - EMC
 - Bound moisture
 - Unbound moisture
- 15) Solid mixing equipment are commonly referred to
- Mixers
 - Blenders
 - Both a & b
 - None
- 16) Impellers are the devices which supply energy form
- Solid-solid mixing
 - Solid-liquid mixing
 - Liquid-liquid mixing
 - Liquid evaporation
- 17) Clarification is a process where amount of solids to be separated from liquid is less than?
- 1%
 - 5%
 - 10 %
 - 0.1 %

- 18) Which one of the following filter is immersed in the slurry to be filtered?
- Sintered glass filter
 - Leaf filter
 - Plate and frame filter
 - Seitz filter
- 19) Which one of the following is a continuous type centrifuge?
- Super centrifuge
 - Perforated basket centrifuge
 - Non-perforated basket centrifuge
 - None of these
- 20) This form of corrosion occurs due to concentration difference in a component?
- Uniform
 - Galvanic
 - Inter-Granular
 - Stress

Q. No.2 Attempt any TWO questions of the following: (20 Marks)

- Explain principle, construction, working mechanism, applications along with diagram of Silversen emulsifier
- Write in detail about Bernoulli's theorem and its application
- Explain principle, construction, working mechanism, applications along with diagram of Rotary drum filter

Q. No.3 Attempt any SEVEN questions of the following: (35 Marks)

- Give mechanism of size reduction, explain impact of speed on ball mill performance.
- Explain construction, working, principle of air separator
- Differentiate between size reduction and size separation
- What is counter flow heat exchangers, explain shell & tube exchanger
- Describe efficiency and capacity of multiple effect evaporators
- Explain mechanism of drying & concept of EMC
- What is top-driven and under driven batch type perforated basket centrifuge
- What is corrosion, give its types and explain how corrosion can be prevented
- Explain principle and procedure of molecular distillation, what are its applications

----- END OF PAPER -----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: III

Subject Name: Pharmaceutical Engineering

Subject Code: BP304T

Max. Marks: 75

Date: 30/07/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

- 1) When principle of conservation of energy is applied to flow of fluids then resulting equation is known as
 - a) Reynold's Number
 - b) Bernoulli's theorem
 - c) Hagen-poiseuille's equation
 - d) Kick's theory
- 2) If the vapour pressure of the liquid is more the evaporation rate is
 - a) High
 - b) Low
 - c) Medium
 - d) Too low
- 3) Mechanism of fluid energy mill is
 - a) Impact pressure
 - b) Attrition and Impact
 - c) Cutting
 - d) None of the above
- 4) Climbing film evaporator also called as
 - a) Falling film evaporator
 - b) Triple effect evaporator
 - c) Rising film evaporator
 - d) Forced circulation
- 5) Which of the following is not an advantage of size reduction?
 - a) Improved dissolution rate
 - b) Improved physical stability
 - c) Improved absorption rate
 - d) Drug degradation
- 6) Which of the following theory not describe rate of filtration?
 - a) Darcy's Law
 - b) Poiseuille's equation
 - c) Kozeny carman equation
 - d) Noye's Whitney equation
- 7) Flow pattern in liquid-liquid mixing
 - a) Radial flow
 - b) Tangential flow
 - c) Longitudinal flow
 - d) All of the above
- 8) Alcohol and water is example of
 - a) Positive mixture
 - b) Negative mixture



- c) Neutral mixture d) None of the above
- 9) Centrifugation is used for...
- a) Mixing b) Purification
- c) Separation d) Sizing
- 10) Which of the following is type of stainless steel?
- a) Martensitic b) Ferritic
- c) Austenitic d) All of the above
- 11) Which of the following is not a filter aid?
- a) Diatomaceous earth b) Perlite
- c) Cellulose d) Cotton
- 12) Austenitic consists of
- a) 13 to 20% chromium +6 to 22% nickel +0.1 to 0.25% carbon
- b) 12 to 20% chromium +2% Nickel+0.2 to 0.4 % carbon
- c) 20 to 40% chromium+12% nickel +1 to 25 carbon
- d) 15 to 30 % chromium+0.15%carbon
- 13) Impingement, enlargement and straining are related to
- a) Mixing b) Centrifugation
- c) Filtration d) All of these
- 14) What is pore size of filtration membrane to remove bacteria?
- a) 0.25 μ m b) 0.27 μ m
- c) 0.22 μ m d) 0.26 μ m
- 15) Mixing device technically called as
- a) Impellers b) Turbines
- c) Paddles d) All of these
- 16) Raoult's law is related to
- a) Vapour pressure b) Atmospheric pressure
- c) Osmotic pressure d) All of the above
- 17) Heat transfer place as per
- a) Zeroth law of thermodynamics b) First law of thermodynamics
- c) Second law of thermodynamics d) Kirchhoff's law
- 18) Corrosion of metal involves
- a) Physical reaction b) Chemical reaction
- c) Both a) and b) d) None of the above
- 19) Free moisture content is
- a) Total water content minus equilibrium moisture content.
- b) Total water content plus equilibrium moisture content
- c) Ratio of total water content to the equilibrium moisture content
- d) Total water present in solid minus water in environment.
- 20) Tunnel dryer is variant of
- a) Rotary drum dryer b) Fluidized bed dryer

c) Tray dryer

d) Spray dryer

Q. 2. Long Answers) = 2 x 10 = 20 (Answer 2 out of 3)

- 1) Discuss in detail various modes of heat transfer. Draw a neat diagram of shell and tube heat exchanger and explain its working.
- 2) Define distillation, write application of distillation and explain construction working laboratory scale vacuum distillation unit.
- 3) Define and classify evaporation and Describe in detail factor affecting evaporation.

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

- 1) Write a note on theories of filtration.
- 2) Explain principle, construction and working of rotameter.
- 3) Explain theories of corrosion.
- 4) Discuss the principle and application of centrifugation.
- 5) Explain in detail multiple effect evaporators.
- 6) Write in detail factors affecting size reduction.
- 7) How will you carry out conveying of solid?
- 8) Write a note on lyophilizer.
- 9) List the equipments used for solid mixing in pharmaceutical industry. explains construction and working of sigma blade mixer.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

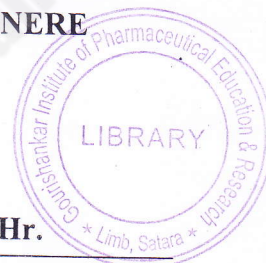
End Semester Examination – Winter 2022

Date:

Course: **B. Pharmacy**
Subject Name: **Pharmaceutical Engineering**
Max Marks: **75**

Sem:
Subject Code:
Duration:

3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) What is Reynold's number? Give its importance.
- ii) What are heat exchangers. Give their types.
- iii) Define evaporation. Classify evaporators.
- iv) Give the applications of drying.
- v) Describe the modes of size reduction.
- vi) Differentiate between solid mixing and liquid mixing.
- vii) Write the advantages and disadvantages of plastics as material of construction.
- viii) Explain the term pitting corrosion and galvanic corrosion.
- ix) Explain the mechanism of filtration.
- x) State Fourier's law with equation.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Define Centrifugation. Classify centrifuges with suitable examples. Discuss in detail on perforated basket centrifuge.
- ii) Write the advantages of size reduction. Discuss the factors affecting selection of a mill for size reduction.
- iii) Classify distillation. Explain the principle, construction, working and applications of molecular distillation.

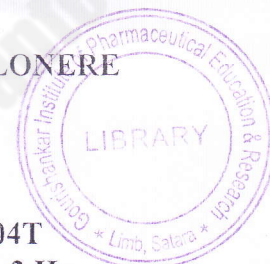
Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Explain in detail about short tube evaporator.
- ii) With the help of neat labelled diagram explain fluidised bed dryer.
- iii) Write the theory of vortex formation and give its prevention methods.
- iv) Discuss on the various modes of size separation.
- v) What are filter aids? Why are they used. Enlist the filter aids used in pharmacy practice.
- vi) Classify materials of construction. Discuss about various types of ferrous metals used.
- vii) Describe the various modes of heat transfer with suitable examples.
- viii) Explain the factors influencing mixing of solids. Write the principle of planetary mixer
- ix) Explain with the help of diagram the construction and working of a Hammer mill

Course: B. Pharmacy
Subject Name: Pharmaceutical Engineering
Max Marks: 75

Sem:3
Subject Code:BP304T
Duration: 3 Hr.

**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) What is Reynold's number? Give its importance.
- ii) What are heat exchangers. Give their types.
- iii) Define evaporation. Classify evaporators.
- iv) Give the applications of drying.
- v) Describe the modes of size reduction.
- vi) Differentiate between solid mixing and liquid mixing.
- vii) Write the advantages and disadvantages of plastics as material of construction.
- viii) Explain the term pitting corrosion and galvanic corrosion.
- ix) Explain the mechanism of filtration.
- x) State Fourier's law with equation.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Define Centrifugation. Classify centrifuges with suitable examples. Discuss in detail on perforated basket centrifuge.
- ii) Write the advantages of size reduction. Discuss the factors affecting selection of a mill for size reduction.
- iii) Classify distillation. Explain the principle, construction, working and applications of molecular distillation.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain in detail about short tube evaporator.
- ii) With the help of neat labelled diagram explain fluidised bed dryer.
- iii) Write the theory of vortex formation and give its prevention methods.
- iv) Discuss on the various modes of size separation.
- v) What are filter aids? Why are they used. Enlist the filter aids used in pharmacy practice.
- vi) Classify materials of construction. Discuss about various types of ferrous metals used.
- vii) Describe the various modes of heat transfer with suitable examples.
- viii) Explain the factors influencing mixing of solids. Write the principle of planetary mixer
- ix) Explain with the help of diagram the construction and working of a Hammer mill.

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2023

Course: B. Pharmacy

Semester: III

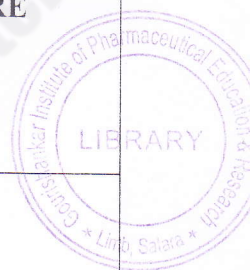
Subject Name: Pharmaceutical Engineering

Subject Code: BP304T

Max Marks: 75

Date: 09/01/2024

Duration: 3 Hr.

**Instructions to the Students:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer All the Questions)	(10 X 2) = 20
i)	What is the mechanism of size reduction?	
ii)	Give applications of filtration	
iii)	What is drying curve?	
iv)	Draw well labelled diagram of leaf filter	
v)	Define corrosion with example	
vi)	Give applications of centrifugation	
vii)	What is seive number?	
viii)	What is filter aid?	
ix)	What is heat exchanger?	
x)	Give applications of evaporation	
Q.2.	Long Answers (Answer 2 out of 3)	(10 X 2) = 20
i)	What do you mean by fluid flow, fluid statics and fluid dynamics? Differentiate between orifice meter and Venturimeter. Describe Venturimeter in detail	
ii)	Define distillation. Explain the principle and working of steam distillation	
iii)	Classify equipments used for mixing of semisolids. Describe the principle, construction and working of ribbon blender	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Explain the Reynold's experiment, give its significance.	
ii)	Explain principle, construction & working of Ball Mill	
iii)	Explain principle, construction and working of cyclone separator	
iv)	Explain principle, Construction, working & uses of planetary mixer.	
v)	Explain principle, Construction working of perforated basket centrifuge.	
vi)	Explain the principle of molecular distillation	
vii)	Explain principle, Construction, working & uses of fluidized bed dryer.	
viii)	Write a note on ferrous metal as material for plant construction.	
ix)	Differentiate between evaporation, distillation and drying explain the factors affecting evaporation	
*** END OF THE PAPER ***		

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: IV

Subject Name: PHARMACEUTICAL ENGINEERING

Subject Code: BP304T

Max Marks: 75

Date: 20-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer All the Questions)	(10 X 2) = 20
i)	In short explain size separation and its importance in pharmacy.	
ii)	Give mechanism of size reduction	
iii)	Draw neat and labelled diagram for plate & frame filter used in filtration process.	
iv)	Define centrifugation. Give applications of centrifugation.	
v)	Ball mill is not useful for size reduction of fibrous material. Explain.	
vi)	What are manometers? What different types of manometers do you know?	
vii)	Give statement for, along with equation, for Fourier's law.	
viii)	What is sieve shaker?	
ix)	Give heat transfer mechanisms	
x)	Give mechanism of size separation	
Q.2.	Long Answers (Answer 2 out of 3)	(10 X 2) = 20
i)	Define distillation. Explain the principle and working of steam distillation	
ii)	Classify equipments used for mixing of semisolids. Describe the principle, construction and working of ribbon blender.	
iii)	What do you mean by fluid flow, fluid statics and fluid dynamics? Differentiate between orifice meter and venturimeter. Describe venturimeter in detail.	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Explain the Reynold's experiment, give its significance.	
ii)	Explain principle, construction, working of perforated basket centrifuge.	
iii)	Explain principle, construction & Working of Ball mill.	
iv)	Explain theory & factors affecting filtration.	
v)	Explain principle, construction & working of cyclone separator.	
vi)	Explain principle, construction, working & uses of fluidized bed dryer.	
vii)	Differentiate between evaporation, distillation and drying. Explain the factors affecting evaporation.	
viii)	Explain the principle of molecular distillation.	
ix)	Explain principle, construction, working & uses of planetary mixer.	

***** END OF THE PAPER *****